Radar Research - Final Summary

Experimental findings, validated results, limitations and next steps | 2 November 2026

Scope. This report summarizes Task 20 using the experiment outputs shared in our conversation and an earlier partial Task 20 research archive. The later 144-file ZIP was not attached here, so this is a concise, evidence-limited summary rather than a line-by-line audit of the final archive.

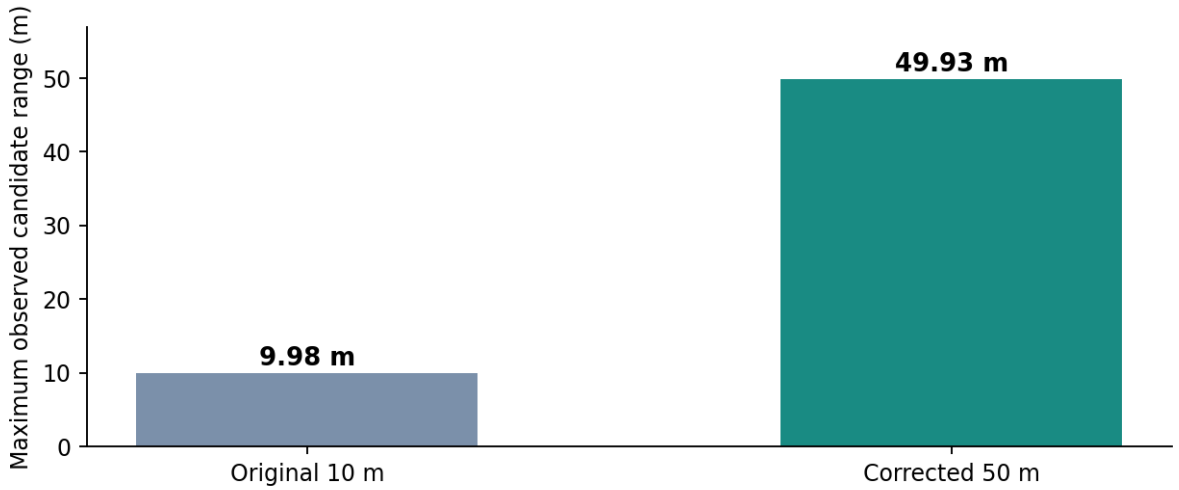
1. Purpose and workflow

Investigate GEMSTAR automotive radar output from Unreal Engine / SUMO, parse recorded OSI SensorView and GroundTruth messages, reconstruct and inspect radar candidate returns, diagnose radar configuration, and evaluate whether ground-truth vehicles can be associated with radar reflections.

Data path: GEMSTAR simulation → recorded .osi messages → Python parsing and geometry analysis → CSV, figures and research notes. A MATLAB radar-visualization script also appears in the earlier archive; a complete MATLAB performance study is not part of the validated final results.

2. Verified engineering outcome

Item	Original configuration	Corrected configuration
Runtime radar range	10.000 m	50.000 m
Maximum measured candidate range	9.9814 m	49.9283 m
Candidate returns in first 100 frames	2,957 (comparison baseline)	5,696
Candidate returns beyond 10 m	0	2,687



Maximum recorded candidate distance. The runs were not synchronized or scene-paired; return-count differences must not be interpreted as a controlled range-only effect.

3. What changed in GEMSTAR

Source investigation found that radar startup initialized a 10 m sensor-view range, while the radar INI setting was not fully propagated. The team extended the radar configuration structure, connected range parsing and assignment, adjusted initialization behavior, and set the radar experiment to **Range=50**. A successful Unreal Editor build and a runtime diagnostic reporting **50.000 metres** supported the corrected setup.

TASK 20 | FINDINGS

4. Data quality and ground-truth alignment

Evidence	Finding
50 m recording	5,696 candidate returns in frames 0-99; 5,614 in frames 1-99
Startup anomaly	50 near-zero candidate returns occurred in frame 0
Timestamp audit	99 exact timestamp matches for SV/GT frames 1-99
External GT observations	112 entries; 3 unique external vehicle IDs
Eligible centre observations	5 inside nominal 50 m range and FoV, all for vehicle ID 76765

Exact timestamp labels do not prove that the underlying sensor trace and object poses were sampled at the same instant. Coordinate-frame assumptions and vehicle bounding-box conventions also remain under investigation.

5. Vehicle and radar-ray experiment

For vehicle **BP_Large_Car_C (ID 76765)** in frames 95-99, the recorded GT centre moved from about 29.84 m to 27.15 m from the radar. Nearest reconstructed radar points were roughly 11.48-13.47 m from the vehicle centre. No point fell within the provisional 3 m proximity threshold.

The closest discrete ray (ray 54) had a centre-to-ray angular separation of 2.04-3.08 degrees and no valid radar return. A ray-to-centre separation analysis produced distances of 1.061-1.460 m.

Geometric assumption	Five-frame result
Recorded GT box dimensions	No predicted ray-box intersections
Doubled GT box dimensions	Two predicted ray-box intersections per frame
Ray 44	Five positive returns at 41.2556 m
Ray 54	Five predicted doubled-box intersections, no positive returns
Predicted doubled-box entry	24.89-27.51 m (different from ray 44 return)

Interpretation: A positive radar reflection on ray 44 cannot be attributed to the target vehicle. The predicted target intersection and actual return are too different, and the actual Unreal collision-mesh hit actor was not recorded. The doubled-box interpretation is a hypothesis, not independently verified truth.

6. Conclusions and limits

Supported: Configurable radar range was implemented and verified in the simulator; recorded candidate returns extended to almost 50 m; reproducible parsing and diagnostics were created.

Not demonstrated: Object-level radar detection accuracy, verified world-coordinate reconstruction, true collision-mesh hit identity, CPU/FPGA acceleration, power measurements, or energy savings. A 0/5 proximity result is **not** a measured 0% vehicle detection accuracy.

7. Research handover and next tasks

Task 21: MATLAB import, radar distribution analysis and reproducible software timing benchmarks. **Tasks 22-23:** Parameterize data loaders for new maps and confirm Camera/LiDAR payloads. **Tasks 24-25:** Implement and validate sensor fusion and CPU baselines. **Task 26 onward:** Integrate ARM-FPGA hardware, accelerate selected preprocessing stages and measure comparable latency, power, energy and perception quality.

Dataset reuse: The algorithms can be reused with new GEMSTAR maps and simulations after paths, target IDs, coordinate conventions and file-schema assumptions are parameterized. Previous numeric results do not transfer automatically to new maps.

8. Archived deliverables

The user-reported Task 20.59 inventory contained **144 files**, including **29 Python scripts, 44 CSV files and 7 images**. Task 20.60 reported zero missing critical files, a saved Markdown report and a saved audit report. Those later files were not present in the partial archive available in this session, so the counts are taken from the shared terminal output.

Expected local report directory: `tests/Task_20_Final_Research/07_Research_Report/`